

Emmanuel Tega Agbragu

Security Engineer

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SUMMARY

Security engineer with 4+ years of experience across offensive and defensive security, specializing in pentesting and the software lifecycle. Expert in manual web and API penetration testing, threat modeling, and developing AI-driven security workflows. Proven track record of identifying critical vulnerabilities and integrating security best practices into the SDLC for regulated banking institutions and research labs.

EXPERIENCE

Security Researcher

Jan 2025 – Apr 2026

The Upanzi Network, Carnegie Mellon University | Kigali, Rwanda

- Performed hands-on pentesting and security code reviews across 5 research platforms, uncovering vulnerabilities and delivering proof-of-concept evidence to ensure remediation before shipping software.
- Chained a mass-assignment flaw with a BOLA/IDOR vulnerability (**CVSS 9.1**) to demonstrate full unauthorized create, read, update, and delete operations on an e-commerce API, quantifying impact for secure design assessment.
- Identified and refuted indirect prompt-injection vulnerabilities (**OWASP LLM01**) in AI-powered features, developing custom agents and workflows to surface AI-specific risks that standard tooling missed.
- Authored NIST-aligned VAPT reports and drove remediation through the SDLC, validating fixes with development teams to ensure security best practices were instilled post-launch.
- Built custom Python tooling and automation scripts to scale endpoint enumeration and debugging across large asset inventories.

Graduate Teaching Assistant

Sep 2024 – Jan 2025

Carnegie Mellon University Africa | Kigali, Rwanda

- Delivered lab sessions in Ethical Hacking and Cybersecurity Operations, reinforcing student competency in offensive security, debugging, and defensive cryptographic algorithms.

Cybersecurity Engineer

Apr 2021 – Jun 2023

Lumenave International Limited | Lagos, Nigeria

- Designed and deployed security infrastructure including IDS/IPS and firewalls for banking clients, utilizing threat modeling to reduce malware infections by **40%**.
- Implemented CyberArk PAM solutions with role-based access controls and least-privilege policies to meet strict regulatory compliance and secure design requirements.
- Configured Symantec DLP policies using cryptographic algorithms and detection rules to prevent unauthorized disclosure and exfiltration of sensitive financial data.
- Led a team of 5 engineers in delivering complex security projects across all deployed solutions.
- Provided technical support and debugging for all security tooling, maintaining **100%** operational efficiency for client environments.

SELECTED PROJECTS

Securing the Digital State: Research Report

2025

Python, Automation | Upanzi Network

- Contributed to a pan-African study assessing the security posture of **21,782** government sub-domains, identifying dangerous exposed methods through automated scanning.
- Built custom automation scripts and tooling powering mass asset discovery, enhancing the ability to discover security issues at scale.

Privileged Access Management Deployment

2021 – 2023

CyberArk, Bash, Oracle, PowerShell | First City Monument Bank

- Deployed a complete CyberArk stack managing privileged account access, integrating secure authentication and least-privilege controls aligned to regulatory SDLC requirements.

TECHNICAL SKILLS

Security Engineering	Burp Suite, OWASP ZAP, Nmap, Nessus, OpenVAS, SQLMap, vulnerability management
Web & Networking	HTTP, TCP/IP, REST APIs, RSA, SAML, OAuth, LDAP, Kerberos, AES, SHA
Programming	Python, Bash, C/C++, Java, SQL, PHP, PowerShell
Frameworks	OWASP, MITRE ATT&CK, NIST, ISO 27001, PCI-DSS
AI & Automation	AI workflows, automation, agentic fluency, LLM security

EDUCATION

MSc Information Technology

Aug 2023 – May 2025

Carnegie Mellon University | GPA: 3.80/4.0 | Mastercard Foundation Partial Scholarship

BSc Pure and Applied Physics, First Class Honours

Mar 2014 – May 2018

University of Benin | Nigeria | GPA: 4.54/5.0 | Top 1% of graduating class

CERTIFICATIONS AND ACHIEVEMENTS

CEH Master • eJPT • AWS Certified Cloud Practitioner • OSCP (In Progress) • Top 10 Finalist, Greenfist CTF 2023 • Mastercard Foundation Scholarship 2023 • First Class Honours, Top 1%, 2018